Project Report

1: INTRODUCTION

1.1 Project Overview

This project provides an in-depth analysis of UNESCO World Heritage sites using Tableau. It visualizes the geographical distribution, preservation categories, and conservation status of 1,121 sites globally.

1.2 Purpose

The purpose of this project is to provide data-driven insights into the current state of global heritage sites, helping to identify trends in preservation and areas at risk.

2. IDEATION PHASE

2.1 Problem Statement

Many UNESCO heritage sites face threats from environmental and human factors. There is a need for an interactive visualization tool to monitor site distribution and identify danger levels effectively.

2.2 Empathy Map Canvas

Users: Researchers, historians, and environmental conservationists.

Needs: To quickly identify endangered sites and understand global distribution patterns.

Challenges: Managing complex, multi-year dataset entries.

2.3 Brainstorming

We focused on using map visualizations for geographical data, Treemaps for danger analysis, and time-series charts to show inscription trends over the last few decades.

3. REQUIREMENT ANALYSIS

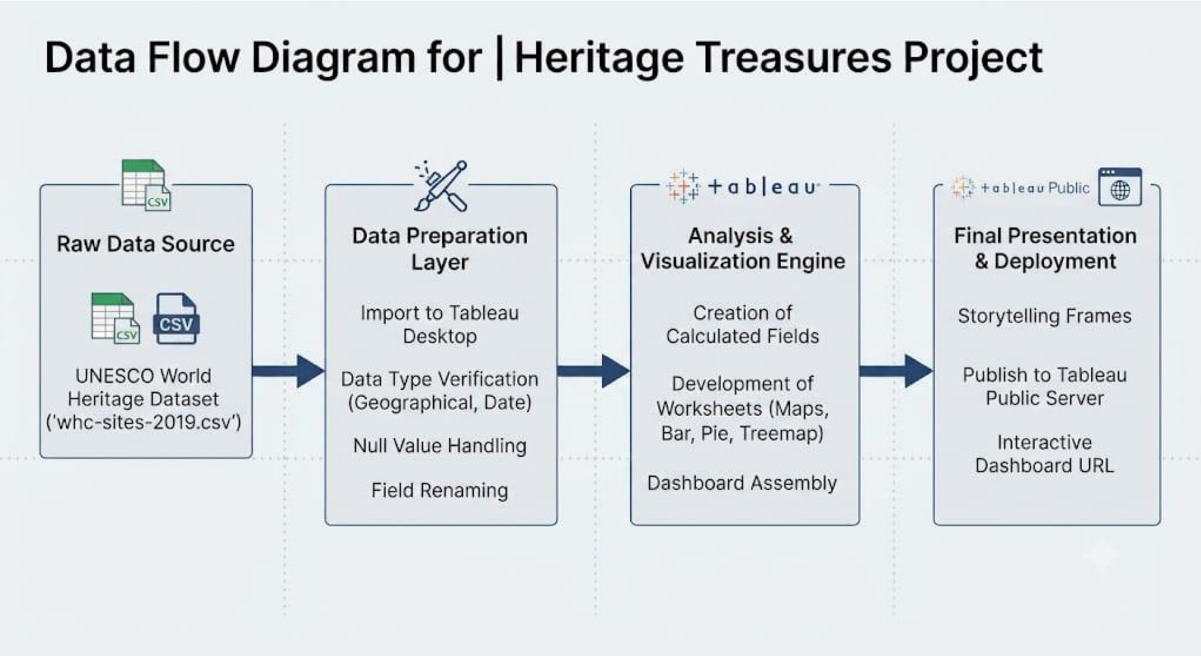
3.1 Customer Journey Map

- Landing: User accesses the dashboard.
- Exploration: User filters data by region or category.
- Insight: User observes trends in danger levels and inscription dates.
- Conclusion: User gains a comprehensive view of global heritage conservation.

3.2 Solution Requirement

Tableau Desktop, UNESCO dataset (whc-sites-2019.csv), and a web browser for Tableau Public.

3.3 Data Flow Diagram



3.4 Technology Stack

Tableau Public, CSV Data.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The interactive dashboard successfully translates raw UNESCO archival data into actionable insights for heritage monitoring.

4.2 Proposed Solution

A multi-view dashboard offering both geographical mapping and categorical analysis to track site preservation.

4.3 Solution Architecture

The architecture follows a three-layer model: Data Source (CSV), Processing Layer (Tableau Calculated Fields), and Presentation Layer (Dashboards and Stories).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase 1 -Data Gathering & Cleaning

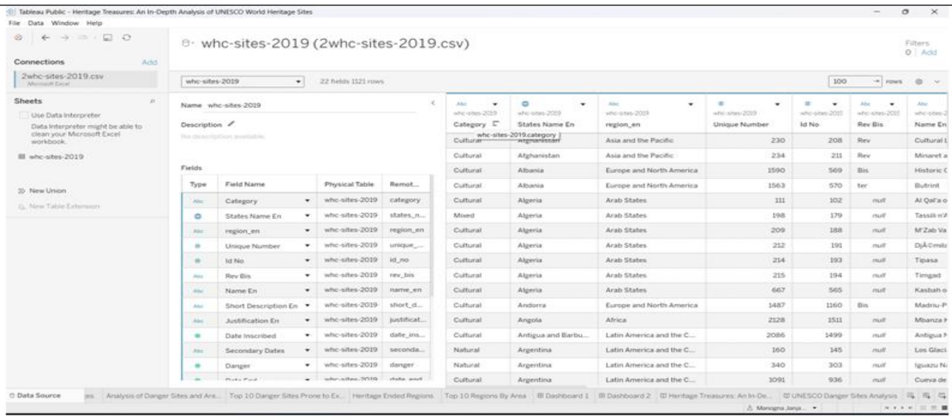
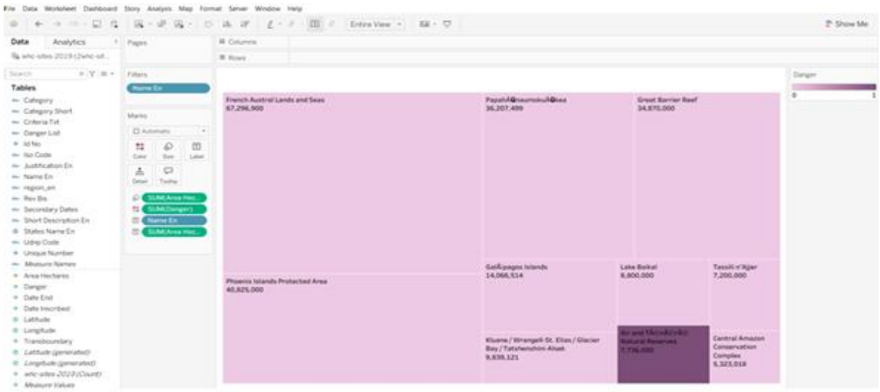
Phase 2 -Dashboard Creation

Phase 3 - Storytelling & Testing

Phase 4 -Deployment to Tableau Public

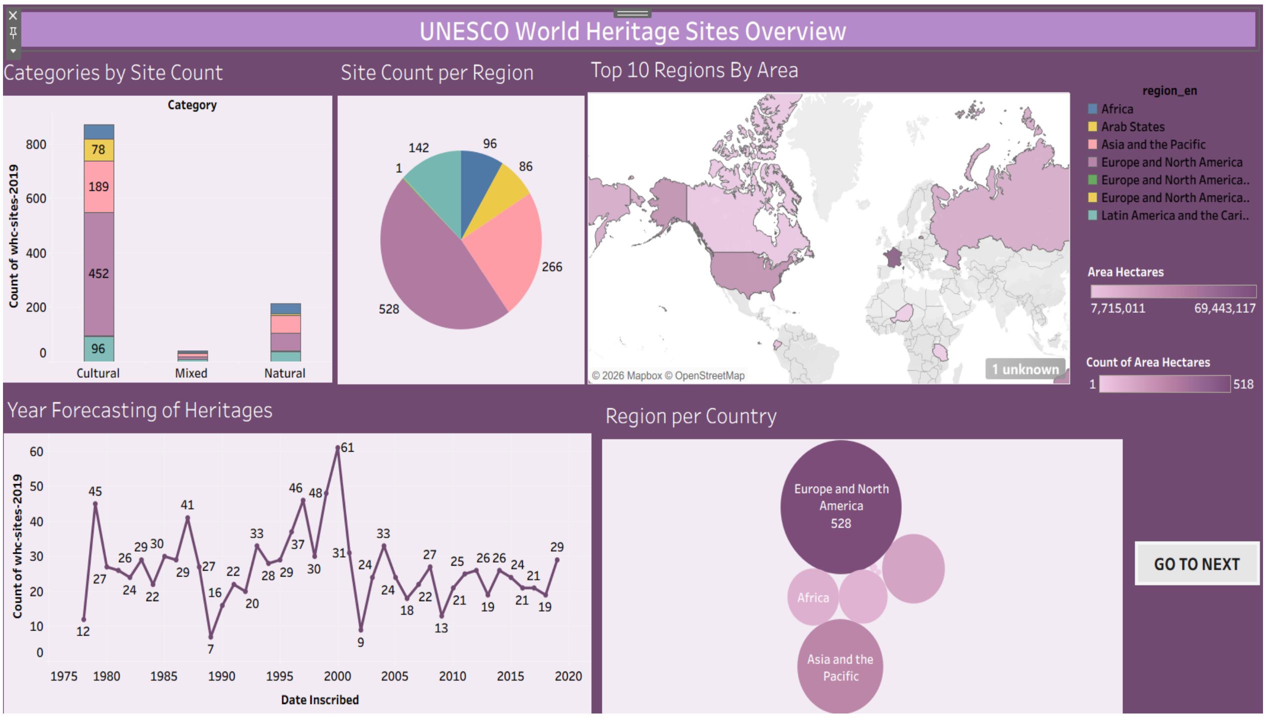
6. FUNCTIONAL AND PERFORMANCE TESTING

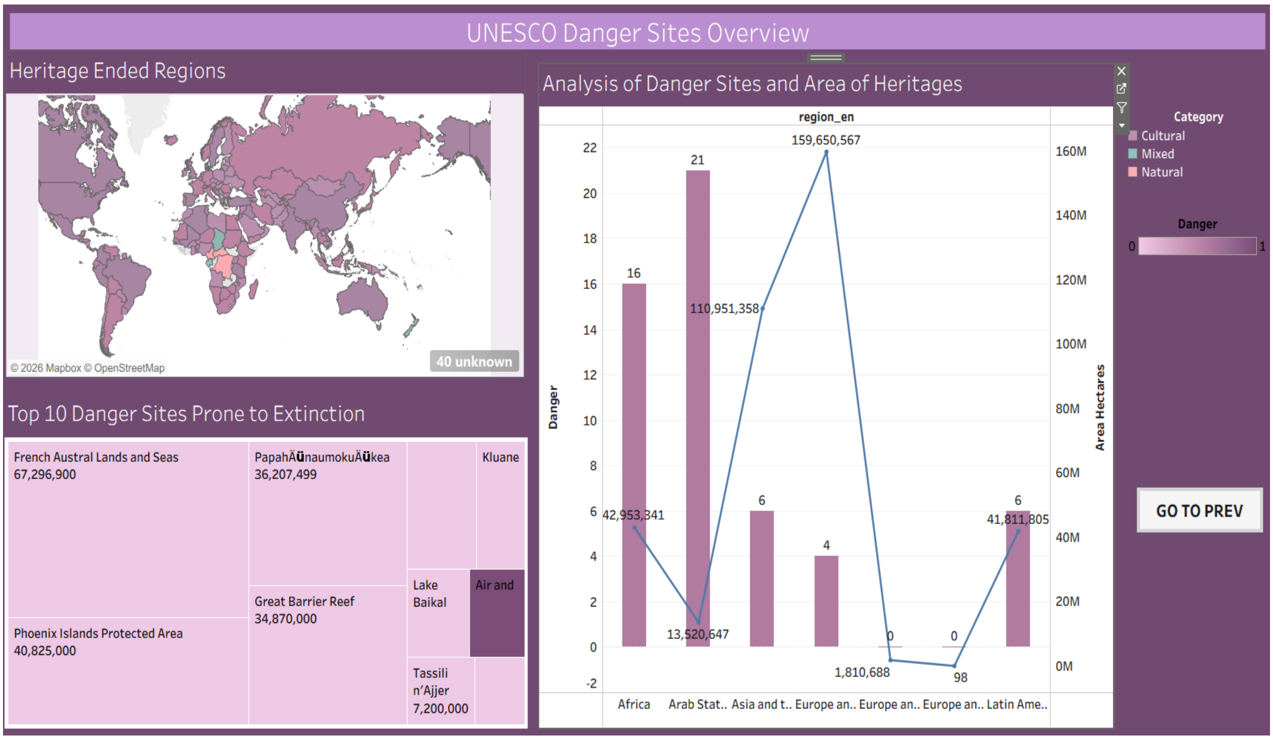
6.1 Performance Testing

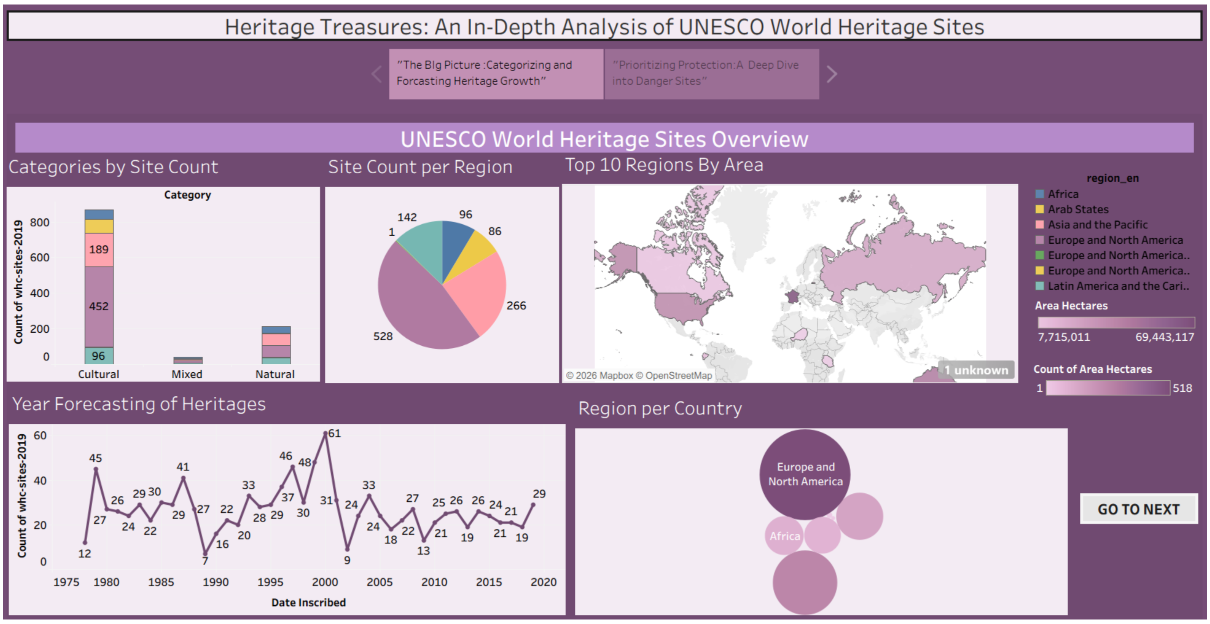
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Preprocessing	Performed data cleaning by removing null values and adjusting filed type to ensure accurate analysis.
3.	Utilization of Filters	 Used the ‘Name En’ field as a filter to refine the data displayed in the visualization.
4.	Calculation fields Used	Used custom calculated fields for data aggregation and metric analysis. Calculation Field Name: Site Count – this calculation uses the COUNT([Name En]) function to aggregate the total number of heritage sites, enabling the dashboard to display the volume of sites per region or category. Calculation Field Name: Heritage Era Group- this calculation uses an IF/THEN logic to group heritage sites based on their inscription year, allowing users to analyse historical preservation trends across different time periods. Calculation Field Name: Endangered Site Percentage- This calculation divides the count of sites marked as ‘Endangered’ by the total count of sites, providing a normalized metric to assess the urgency of preservation efforts.
5.	Dashboard design	No of Visualizations / Graphs - 8
6	Story Design	No of Visualizations / Graphs -2

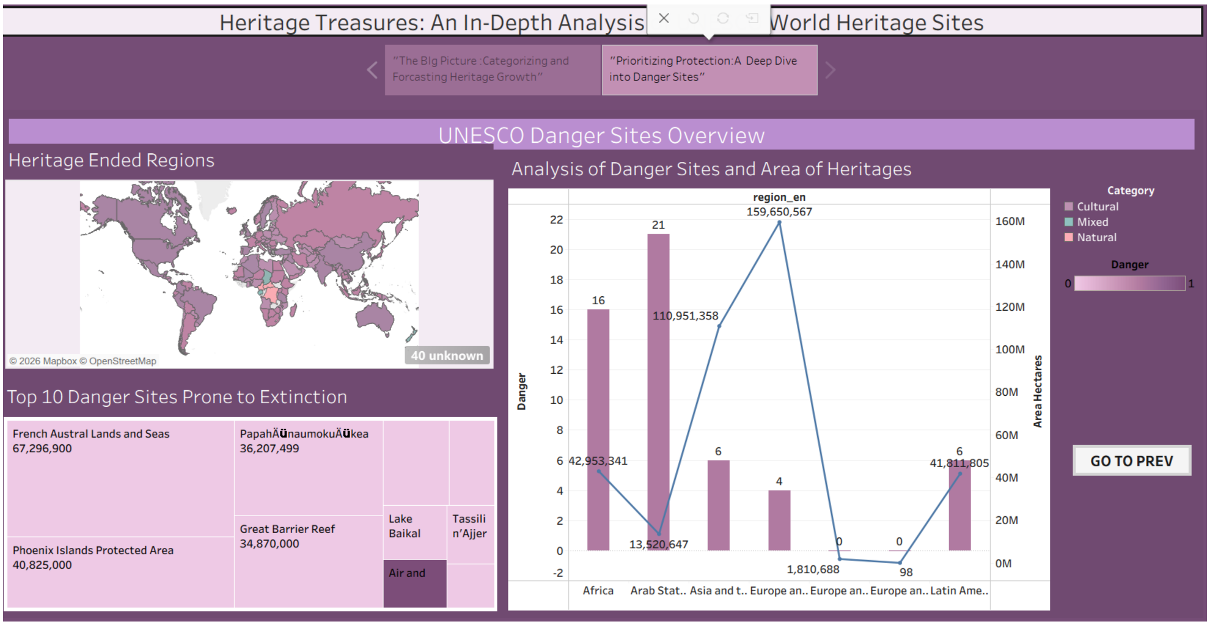
7. RESULTS

7.1 Output Screenshots









8. ADVANTAGES & DISADVANTAGES

Advantages: Highly interactive, visually intuitive, and allows for real-time data filtering.

Disadvantages: Dependent on the periodic updates of the UNESCO dataset; limited by the availability of granular historical data.

9. CONCLUSION

The "Heritage Treasures" project effectively visualizes global heritage data, providing a clear understanding of the distribution, categories, and safety status of UNESCO sites.

10. FUTURE SCOPE

Future iterations could integrate real-time climate data to track how environmental changes impact site conservation status globally.

11. APPENDIX

Source Code:

Tableau Calculated Fields (No traditional source code).

Dataset Link:

https://www.kaggle.com/datasets/ujwalkandi/unesco-world-heritage-sites/data?select=whc-sites-2019.csv

GitHub & Project Demo Link:

Project Demo Link (Tableau Public URL)-Profile - manogna.janjanam | Tableau Public

https://public.tableau.com/app/profile/manogna.janjanam/vizzes (both URLs are same)

GitHub - https://github.com/Manogna322